

Saturation Temperature and Pressure Lab

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Synopsis

The relationship between the saturation temperature and absolute pressure of water was investigated using an Armfield TH3 saturation-pressure apparatus. Nine equilibrium-stage readings were recorded from 50.00 kPa to 450.00 kPa gauge pressure. The measured atmospheric pressure of 101.65 kPa was added to each gauge reading. The original PT100 resistance readings were converted using the two-point relationship recorded on the laboratory whiteboard for this session: 109.0 Ω at 15 °C and 138.7 Ω at 100 °C.

Both the experimental and published steam-table data followed a smooth power relationship of the form $T = AP^n$. The experimental fit was $T = 37.633P^{0.2146}$, compared with $T = 31.019P^{0.2557}$ for the direct steam-table values. Experimental temperatures were below point-specific interpolated theoretical values at all nine pressures. The mean signed error was -4.32% , and the largest absolute error was 6.47% . The experiment reproduced the expected increasing saturation trend, but the growing negative deviation indicates a systematic influence. Plausible contributors include thermal lag, heat loss, residual air and uncertainty introduced by the session-supplied linear conversion. Repeated readings at each pressure would be required to separate repeatability from systematic bias.

1 Aim

To measure the relationship between the saturation temperature and absolute pressure of water, determine an empirical power-law model, and assess the accuracy of the apparatus by comparison with published saturated-water data.

2 Theory and Data Processing

At liquid–vapour equilibrium, saturated liquid water and saturated steam have one saturation temperature for a given absolute pressure. Increasing pressure raises this temperature and produces a smooth, nonlinear saturation curve. The electronic pressure reading was gauge pressure, therefore

$$P_{\text{abs}} = P_1 + P_{\text{atm}}, \quad P_{\text{atm}} = 101.65 \text{ kPa.} \quad (1)$$

The laboratory whiteboard explicitly supplied two resistance–temperature points for processing this session’s readings:

$$\alpha = \frac{R_2 - R_1}{T_2 - T_1} = \frac{138.7 - 109.0}{100 - 15} = 0.34941 \text{ } \Omega/\text{ }^\circ\text{C.} \quad (2)$$

The experimental temperature was therefore calculated as

$$T_{\text{exp}} = T_1 + \frac{R - R_1}{\alpha} = 15 + \frac{R - 109.0}{0.34941}. \quad (3)$$

This is the method documented in the primary evidence from the actual laboratory session. It differs from the manufacturer’s general Data Sheet 1 bridge correction followed by the Data Sheet 2 PT100 reference chart. The session relationship was retained so the report reproduces the experiment as

instructed; the difference from the manual is treated as a methodological limitation rather than concealed as an apparatus calibration.

Over the measured range, both datasets were represented by

$$T_{\text{sat}} = AP_{\text{sat}}^n, \quad (4)$$

where A and n were obtained by a log-space least-squares fit. Direct values from saturated-water pressure Table A-5 were used for the comparison curve. For the error calculation only, the theoretical temperature at each experimental pressure was found by linear interpolation:

$$T_{\text{th}} = T_a + \frac{P_{\text{abs}} - P_a}{P_b - P_a}(T_b - T_a). \quad (5)$$

The signed percentage error was

$$\text{Error (\%)} = \frac{T_{\text{exp}} - T_{\text{th}}}{T_{\text{th}}} \times 100. \quad (6)$$

3 Equipment

The equipment comprised an Armfield TH3 saturation-pressure apparatus with a 2.40 L boiler, sight glass, closed pipe loop, cartridge heaters, control console, filling and pressure-relief valves, a PT100 platinum resistance thermometer, electronic and Bourdon pressure gauges, a barometer, and de-ionised water. The manual gives a normal operating range of 0–7.00 bar gauge and a maximum allowable pressure of 8.00 bar gauge.



Figure 1: Armfield TH3 boiler and pipe loop used in the 2026 experiment. The photograph shows the boiler, pressure gauge, viewing port, valves and sensor wiring of the apparatus used.

4 Method

1. With the drain and calorimeter isolating valves closed and console power off, the boiler was filled with de-ionised water to halfway up the sight glass.

2. The water was heated to vigorous boiling with the filling point open. Steam was vented until the temperature indication stabilised, allowing air to be expelled from the loop.
3. The filling valve was closed. The apparatus was then treated as a sealed, approximately constant-volume system containing a saturated water–steam mixture. Heater input was raised for approximately two minutes and reduced.
4. Gauge pressure and PT100 resistance were recorded after the indication stabilised at nine stages from 50.00 kPa to 450.00 kPa gauge pressure.
5. Gauge readings were converted to absolute pressure. Each recorded resistance was converted to temperature using the session-supplied two-point relationship documented on the laboratory whiteboard.
6. Experimental points and direct Table A–5 values were plotted on the same axes and fitted separately. Theoretical temperatures were then interpolated at the nine experimental pressures for the point-by-point error calculation.
7. After the final reading, the heaters were switched off and the isolating valve was opened before cooling to prevent a damaging partial vacuum.

5 Results

5.1 Worked calculation

For the first reading, $P_1 = 50.00$ kPa and $R = 142.30 \Omega$:

$$P_{\text{abs}} = 50.00 + 101.65 = 151.65 \text{ kPa}, \quad (7)$$

$$T_{\text{exp}} = 15 + \frac{142.3 - 109.0}{0.34941} = 110.303^\circ\text{C}, \quad (8)$$

$$T_{\text{th}} = 111.35 + \frac{151.65 - 150}{175 - 150}(116.04 - 111.35) = 111.660^\circ\text{C}, \quad (9)$$

$$\text{Error} = \frac{110.303 - 111.660}{111.660} \times 100 = -1.215\%. \quad (10)$$

5.2 Processed results

Table 1: Original measurements, session-derived temperatures, theoretical temperatures and errors.

P_1 (kPa)	P_{abs} (kPa)	R (Ω)	T_{exp} ($^\circ\text{C}$)	T_{th} ($^\circ\text{C}$)	Error (%)
50	151.65	142.3	110.30	111.66	−1.21
100	201.65	144.8	117.46	120.46	−2.49
150	251.65	146.9	123.47	127.62	−3.25
200	301.65	148.7	128.62	133.70	−3.80
250	351.65	150.1	132.63	139.02	−4.60
300	401.65	151.1	135.49	143.75	−5.75
350	451.65	152.6	139.78	148.03	−5.57
400	501.65	153.8	143.22	151.95	−5.75
450	551.65	154.6	145.51	155.57	−6.47

5.3 Saturation curves and fitted equations

Figure 2 compares the processed experimental data with the direct pressure–temperature pairs from Table A–5. Interpolated theoretical points were not used to build either trendline.

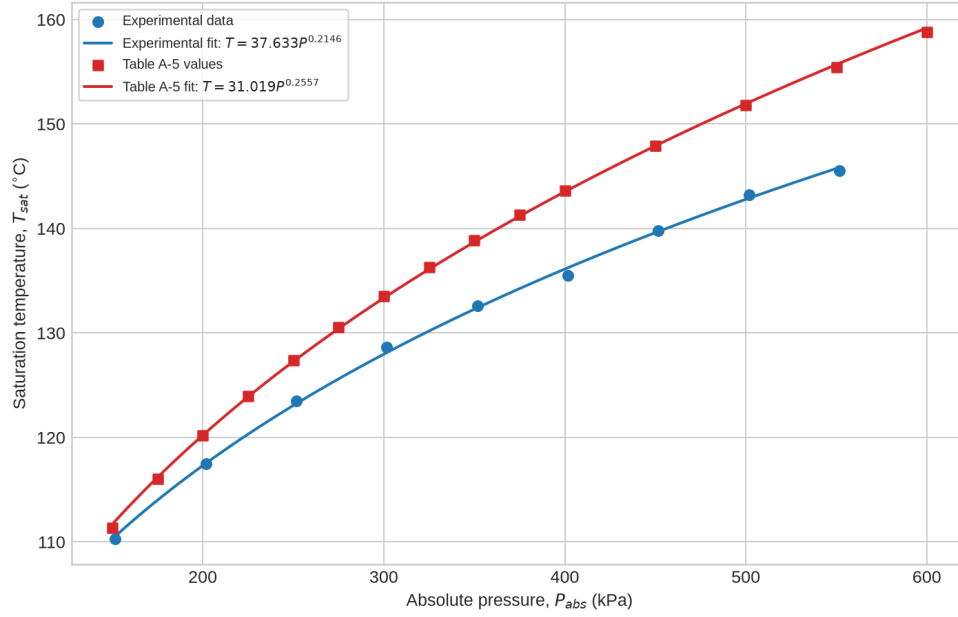


Figure 2: Experimental temperatures and direct Table A–5 saturation values plotted against absolute pressure with separate power-law fits.

Table 2: Power-law fit parameters for $T = AP^n$.

Dataset	A	n	R^2
Experimental measurements	37.633	0.2146	0.99895
Direct Table A–5 values	31.019	0.2557	0.99983

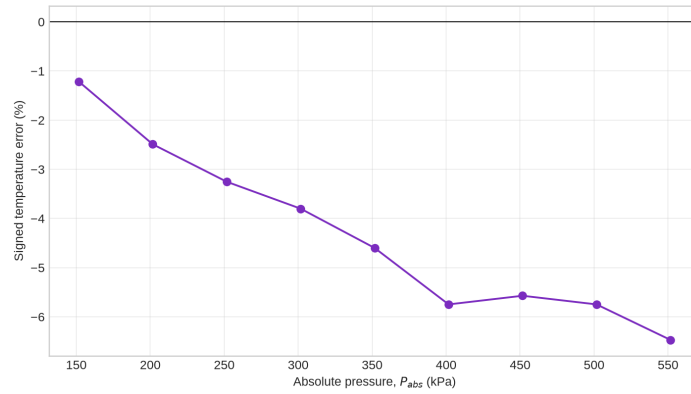


Figure 3: Signed experimental-temperature error relative to the point-specific interpolated saturation temperatures.

The mean signed error was -4.32% . Since every value was below the reference, the mean absolute error was also 4.32% . The maximum absolute error was 6.47% at 551.65 kPa absolute.

6 Discussion

6.1 Comparison and usefulness of the fitted equation

Both curves increased smoothly with absolute pressure, confirming the expected nonlinear relationship. However, the experimental exponent of 0.2146 was lower than the direct-table value of 0.2557 .

The experimental curve therefore predicted less temperature rise with pressure than the reference curve. Both regressions still had $R^2 > 0.998$, showing that the mismatch was systematic rather than randomly scattered.

The experimental equation is useful as a compact empirical description of these measurements between 151.65 kPa and 551.65 kPa absolute. It should not replace steam tables in design work or be extrapolated outside the tested interval because it depends on the session-supplied conversion, measurement procedure and chosen fitting method.

6.2 Error trend

All signed errors were negative, indicating a systematic tendency for the derived experimental temperature to lie below the reference saturation temperature. Error magnitude generally increased with pressure from 1.21% to 6.47%, with a small local reduction at 451.65 kPa. The trend is consistent with a pressure-dependent bias, but the experiment alone cannot identify whether its principal cause was thermal behaviour, residual air, instrument response or the session-supplied resistance conversion.

6.3 Steam temperature, water temperature and assumptions

Saturated liquid and vapour have the same temperature only at thermodynamic equilibrium. The PT100 measured steam in the upper pipework rather than liquid inside the boiler. The comparison therefore made the following assumptions:

- air was sufficiently purged for total measured pressure to approximate the steam saturation pressure;
- pressure, boiler water, local steam and the PT100 had reached a common equilibrium value before each reading;
- atmospheric pressure remained effectively constant during the run;
- the whiteboard values 109.0 Ω at 15 °C and 138.7 Ω at 100 °C were the intended processing points for this session; and
- a straight line between those points adequately represented the recorded resistance range, while steam-table interpolation was adequate over the small pressure intervals used.

These assumptions are reasonable for processing the data, but the experiment did not test each one independently. Consequently, the possible causes below are interpretations consistent with the observed errors, not proven mechanisms.

Thermal lag occurs because energy must conduct through the fluid, pipework and probe sheath before the sensor reaches the local steam temperature. Heat loss from exposed pipework can also keep the probe region slightly cooler than the boiler. Either effect would produce negative temperature error if a reading were taken before complete stabilisation.

6.4 Possible sources of error and improvements

- **Residual air:** remaining non-condensable air contributes partial pressure, so the measured total pressure can exceed the steam partial pressure associated with the local saturation temperature.
- **Thermal lag and heat loss:** sensor response time and temperature gradients can make the upper-pipe measurement lower than the equilibrium boiler temperature.

- **Calibration uncertainty and method sensitivity:** two defensible temperature-processing routes are available. The session whiteboard gives a two-point relation, whereas the Armfield manual specifies Data Sheet 1 bridge correction followed by Data Sheet 2 conversion. Every recorded resistance (142.3–154.6 Ω) was above the whiteboard’s upper point of 138.7 Ω at 100 °C. The session calculation was therefore an extrapolation, not an interpolation, and assumed that one constant slope remained valid above 100 °C. The available evidence also does not establish whether both whiteboard endpoint resistances were intended as indicated or bridge-corrected values; the manual distinguishes between these quantities. Endpoint rounding, display resolution and uncertainty in the stated 15 °C reference would consequently affect every derived temperature.

Applied to the same raw readings, the session method gives a mean signed error of -4.32% , whereas the manual method gives -1.03% . The manual temperatures are higher by approximately 0.75 °C at the first point and 8.32 °C at the final point. This widening difference demonstrates that conversion choice can create a pressure-dependent error trend. These two results are sensitivity cases, not statistical uncertainty limits; without a traceable calibration and repeated readings, a defensible confidence interval cannot be assigned. This report retains the session method because it is documented in the primary evidence from the actual laboratory session.

- **Pressure measurement:** pressure-sensor calibration, display resolution and atmospheric pressure uncertainty affect absolute pressure.
- **Single readings:** one value at each stage cannot distinguish random variation, hysteresis and repeatability from systematic bias.

The experiment would be stronger if each pressure stage were repeated after independent stabilisation. Means and standard deviations could then be reported, and an increasing-pressure run could be compared with a decreasing-pressure run to identify hysteresis. Longer air purging, an explicit stability criterion for both pressure and resistance, and documented instrument resolutions would further strengthen the uncertainty assessment.

7 Conclusion

The experiment reproduced the expected nonlinear increase in water saturation temperature with absolute pressure. Using the two-point resistance conversion recorded for this laboratory session, the experimental fit was $T = 37.633P^{0.2146}$ with $R^2 = 0.99895$, compared with the direct steam-table fit $T = 31.019P^{0.2557}$. Experimental temperatures were below point-specific theoretical values at all nine stages, with a mean signed error of -4.32% and a maximum absolute error of 6.47%. The increasing error magnitude indicates a systematic influence but does not prove a single cause. Thermal lag, heat loss, residual air, pressure uncertainty and the difference between the session method and the manufacturer’s Data Sheet procedure remain plausible contributors. Repeated measurements are recommended to quantify repeatability and help separate experimental behaviour from conversion-method bias.

References

1. Y. A. Çengel, M. A. Boles, and M. Kanoğlu, *Thermodynamics: An Engineering Approach*, 9th SI ed., McGraw-Hill, 2020, Table A–5.
2. Armfield Ltd., *TH3 Saturation Pressure Apparatus Instruction Manual*, Issue 15, Data Sheets 1–2, pp. 22–24.
3. Auckland University of Technology, *ENME 601 Saturation Temperature & Pressure Lab Instructions*, 2025 update used for the 2026 laboratory.
4. Auckland University of Technology, laboratory whiteboard and TH3 controller photographs, Tuesday 8–10 session, 2026.